

RESILIENT FOG-BASED ADAPTIVE TRAFFIC CONTROL SYSTEM

Ashwini Matange*, Varun Taneja†, Aarya Chaumal‡, Pallavi Buwa§ and Jibi Abraham¶

Dept. of Computer Engineering and Information Technology, COEP Tech

Pune, India

Email: *asm.comp@coep.ac.in, †tanejavp19.comp@coep.ac.in, ‡chaumalam19.comp@coep.ac.in,
§buwapu19.comp@coep.ac.in, ¶ja.comp@coep.ac.in

Abstract—Traffic congestion is a growing problem in India, largely caused by the increasing number of vehicles. To address this, creating an adaptive traffic control system has become essential. To present a solution, this paper aims to create a resilient, adaptive traffic control system that dynamically adjusts signal timings based on the current traffic density. The paper proposes introducing a Fog layer at each traffic junction to capture real-time video feeds and process them locally, with the final decision about dynamic signal control being taken in the Cloud. The validity of the solution is tested through simulations in SUMO. A Machine Learning model is trained in the Cloud to predict expected vehicle numbers to set adaptive signal times for contingency situations. The paper aims to demonstrate the resilience and performance of the system with the adaptive traffic signal control algorithm, potentially offering a solution to traffic congestion in India.

Index Terms—Fog computing, Cloud computing, Traffic control, Computer vision, Simulation

I. INTRODUCTION

With a population of more than 1.3 billion, India is one of the most populated nations in the world. India's population is expanding quickly, which has resulted in a large rise in the number of vehicles on its roadways, which has caused severe traffic congestion in many of the nation's cities. The main characteristics of heavy traffic on the roads include slower speeds, longer travel times, and an increase in vehicle queuing. Several Indian towns are plagued by this serious issue. Mumbai, Delhi, and Bangalore have routinely ranked among the top 10 most crowded cities in the world according to several traffic indices during the past few years [1].

Traffic congestion is a pressing problem that needs to be dealt with as soon as possible due to its adverse effects on overall health and well-being. The increase in the number of vehicles on the road has led to longer travel times, increased pollution, increased physical and mental issues along with significant economic costs to the country. To combat this issue in a quick and efficient manner, adaptive traffic signals offer a promising solution.

Adaptive traffic control systems gather and interpret data regarding traffic volume, speed, and congestion levels at junctions. They use a mix of sensors and cameras to collect the required data and process it to acquire information about the current traffic situation. Algorithms use this extracted data to analyze traffic patterns and modify the timing of the traffic

signals such that the lanes with maximum density are given priority. These systems adjust dynamically to changing traffic needs and optimize traffic signal timings to minimize delays and maximize traffic flow by taking into account the present traffic situation and projecting future traffic patterns.

Traditional traffic signal control systems use central servers or purely Cloud-based architectures to process traffic data and make decisions about signal timings. However, these systems may suffer from delays in data transmission and processing. Large amounts of data need to be transmitted to the central server and then stored for processing. This requires massive computing and storage power and leads to heavy bandwidth consumption. This can result in sub-optimal traffic signal timings and reduced effectiveness in managing traffic flow.

In contrast to this, Fog computing leverages distributed computing resources, such as edge devices and local servers that are placed near the traffic junction, to process traffic data in real-time. Distributed processing drastically brings down bandwidth consumption while simultaneously reducing latency. The processing of video feeds happens locally which also reduces storage requirements. This allows for faster data processing, decision-making, and response times, which are critical for adaptive traffic signal control [2].

In this paper, we propose a resilient Fog-based adaptive traffic control system that is ideal for the Indian scenario. The proposed system utilizes data available from the installed CCTV cameras and processes the videos in a distributed fashion in Fog nodes. The final signal cycle time adjustment code runs in the Cloud and takes a decision regarding the time adjustment. Additionally, an ML model is trained using historical data to estimate the number of vehicles for a particular day and junction. These predicted values are used to set signal times in case of Fog node failure. This creates an efficient, resilient, and quick response system that can adapt to changing traffic conditions quickly.

The rest of the paper is organized as follows: Section II introduces the topic of Fog computing. Section III offers a survey of the work done in the domain of using Fog computing for adaptive traffic control. Section IV talks about the advantages of using Fog computing for this purpose. Section V introduces the proposed solution and Section VI elaborates on the results obtained from our work. Section VII offers concluding remarks with Section VIII enumerating the

future scope of the paper.

II. FOG COMPUTING

Fog computing aims to bridge the gap between edge computing and standard Cloud computing. It is a decentralized computing architecture that places network, processing, and storage resources nearer to where data is created and consumed, enabling quicker reaction times, lower latency, and fewer bandwidth needs.

The concept of Fog computing is based on the idea of creating a hierarchy of computing resources that extends from the Cloud down to the edge devices. The Cloud, which offers centralized processing and storage capabilities, is at the top of the hierarchy. The edge devices such as sensors, mobile phones, and Internet of Things (IoT) devices, are at the bottom of the hierarchy. Fog computing is still in its fledgling stage as it is a relatively new concept that is still being explored and developed. Many researchers and industry experts are still exploring its capabilities and limitations. It involves a complex network of interconnected devices and can be considered interdisciplinary in nature while still lacking standardization in hardware, software, and protocols [3], [4].

To explore the concept of Fog computing, Habibi et al. provide a comprehensive survey of the architectural components of the same – Fog nodes, computing infrastructure, services, and applications. They present a taxonomy of Fog computing dimensions and aspects while comparing it with Cloud, mobile edge, and mobile Cloud computing. They explain in detail its use cases, challenges, and limitations [5].

In similar work, Bermbach et al. provide an overview of Fog computing, its challenges, and potential solutions such as an introduction to managed Fog infrastructure services and designing fine-grained topology abstractions for Fog applications. The paper also mentions exploring physical security solutions like tamper-proofing Fog nodes. Overall, it provides valuable insights into the current state of Fog computing and highlights areas that require further research and development [6].

III. RELATED WORK

This section discusses the work done in the creation of an adaptive traffic signal system using the principles of Fog computing along with the techniques used for estimating traffic density and finding patterns in the same.

Liu et al. proposed a secure intelligent traffic light control system using Fog computing. The system architecture consists of three layers: Perception, Fog, and Cloud layer. The proposed system utilizes blockchain technology for secure data transmission and storage, and an identity-based encryption scheme for secure communication between the Fog layer and the Cloud layer [7].

Ha et al. proposed a Fog-Cloud-based multi-agent reinforcement learning (MC-RL) approach for traffic signal control. The proposed approach involves multiple traffic agents and Fog nodes, which communicate with each other and the Cloud to learn optimal traffic signal control policies. The approach uses

a neural network-based value function approximation to model the state-action values and facilitate learning. The results of experiments using a traffic simulation show that the proposed approach outperforms existing traffic signal control methods in terms of traffic throughput and congestion reduction [8].

Hossan et al. proposed a Fog-based dynamic traffic light control system that aims to improve public transport by reducing delays and travel times. The system uses sensors and cameras installed on public transport vehicles to collect real-time traffic information and transmit it to Fog nodes where they process the data and dynamically adjust traffic light timings to prioritize the passage of vehicles used for public transport. The proposed system can significantly cut travel times for public transportation vehicles while maintaining a reasonable level of service for other vehicles on the road [9].

Zeinaly, Zahra, Mahdi Sojoodi, and Sadegh Bolouki proposed a resilient intelligent traffic signal control scheme for intersections to mitigate the effect of accidents on traffic flow using deep reinforcement learning. The scheme consists of the accident detection module and the intelligent traffic signal control module. The accident detection module uses an image processing algorithm to detect accidents at intersections. In the case of an accident, the intelligent traffic signal control module uses deep reinforcement learning to adaptively change the traffic signal timings [10].

Razali et al. provided an overview of traffic flow prediction using machine learning and deep learning techniques. The paper compares the performance of various deep learning and machine learning algorithms, including Support Vector Machines (SVM), Random Forest (RF), Artificial Neural Networks (ANN), and Long Short-Term Memory (LSTM), and discusses the difficulties associated with predicting traffic flow. The performance of these algorithms is also assessed in the paper using a variety of evaluation metrics, including Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). The results demonstrate that deep learning algorithms, like LSTM, outperform conventional machine learning algorithms for predicting traffic flow, but the performance is reliant on the caliber and volume of training data [11].

Deekshetha et al. presented a machine learning-based method for forecasting future traffic using past traffic data. The paper describes the conventional techniques for predicting traffic as well as their shortcomings. The suggested model is based on the decision tree algorithm and forecasts traffic flow using information from weather sensors, traffic sensors, and public transportation schedules. In order to show how successfully the suggested approach predicts traffic flow, the authors also provide the findings of experiments performed on actual traffic data [12].

While our work draws inspiration from that done in the past, we aim to create a solution that is feasible to be implemented in an Indian scenario. Hence, we keep in mind the ease of implementation with the available infrastructure requiring minimum overhead. The approach of using VANETS by Liu et al. [7] is infeasible in a country like India that has seen limited technological advancement. The approach suggested

by Ha et al. [8] of using deep reinforcement learning is computationally expensive and would cause a heavy overhead in terms of hardware. Hossan et al. [9] focus on creating a system to give priority to public vehicles, which cannot be the only objective for a typical Indian city. The unique feature of this work is the creation of a resilient Fog-based system. Zeinaly et al. [10] attempt to create an intelligent traffic system with a focus on congestion mitigation during accidents, which, although critical, does not provide a solution to the issue of general traffic congestion. Considering the work done by Razali et al. [11] and Deekshetha et al. [12], we use machine learning techniques to make predictions about traffic conditions to enable our system to function even when one or more Fog nodes are down.

IV. ADVANTAGES OF FOG COMPUTING FOR ADAPTIVE TRAFFIC CONTROL

The following points highlight the main reasons for implementing an adaptive traffic control system using the Fog computing paradigm.

- 1) Latency and Response Time: Instead of sending terabytes of real-time data captured by traffic cameras to the Cloud for processing, it's sent to Fog nodes situated at intersections. This reduces the round trip time and latency and increases the response times.
- 2) Decentralisation: One node failure won't affect the nodes at other junctions. This would make the proposed solution more resilient during network failures or malfunctions in the hardware.
- 3) Data privacy: As the video processing is done in the Fog nodes, no video containing public information like vehicle number plates, timings of travel, footage of accidents, etc is not sent to the Cloud for processing immediately. This could reduce the chances of hacking and snooping during transmission.
- 4) Operational costs: Processing and transferring thousands of terabytes of data daily would require huge amounts of bandwidth in the Cloud. This can potentially be decreased or minimized by the use of a Fog layer.

V. PROPOSED SOLUTION AND IMPLEMENTATION

A. Adaptive Traffic Signal Control

To create a system that dynamically adjusts signal times based on traffic density, it is proposed to use a Fog plus Cloud architecture. The computation work is divided between the Fog and the Cloud to provide distributed processing, enabling a quick response. This section describes how such a Fog-based Adaptive Traffic Control System can be made. The component diagram of the system is shown in Fig. 1.

The data that has been used in the paper is in the form of videos which are real-time traffic camera feeds from three junctions. The program to identify and count vehicles runs in the Fog nodes. It uses the OpenCV library along with the YOLO library for object detection [13]. The cv2.dnn is a module in the OpenCV that provides an implementation of deep neural networks (DNNs) and is used in the code.

TABLE I: System Specifications

Component	Value
GPUs	4x Tesla V100
GPU Memory	128 GB Total System
CPU	Intel Xeon E5-2698 v4 2.2 GHz (20-Core)
System Memory	256 GB RDIMM DDR4
Storage	2TB SSD
Software	Ubuntu Linux OS DGX Recommended GPU Driver Python with required libraries

For object tracking, the video is divided into frames where with the help of NMS (Non-Maximum Suppression) Boxes, the redundant overlapping bounding boxes for vehicles are removed and the most precise ones are found. This helps by lowering false positives and enhancing the programs' overall accuracy. The output is a CSV file that contains the vehicular count for each type of vehicle in all directions on a particular time stamp. This CSV file is sent to SUMO for the creation of the simulation. System specifications for the setup used for video processing and creation of the dataset is as show in Table I.

The scenarios for the entire duration for which the data is available are created using SUMO [14]. SUMO, which stands for "Simulation of Urban Mobility," is an open-source microscopic traffic simulator designed to model and analyze urban traffic networks. The scenario creation in SUMO consists of creating a network using a network editor like Netedit, defining vehicle routes using a route editor like Duanrouter, generating traffic using a traffic generator like JTRROUTER, and configuring the simulation. A visual of the SUMO simulation is shown in Fig. 2.

The algorithm to find optimal signal times runs in the Cloud. It calculates these by dividing the green signal time in the ratio of the traffic densities on all sides of a junction. This value is constrained by predefined minimum and maximum timing values. The Fog nodes send the count of vehicles to the Cloud which then runs this decision-making algorithm and returns the new signal times. The traffic signal then operates with these modified timings that are better suited to the current traffic situation. The pseudocode for the same is shown in Algorithm 1.

As it is not possible to verify the working of this system on real traffic lights, a part of the city of Pune, India is chosen as the target. The real-life data obtained from CCTV cameras are used to create a simulation in SUMO and the validity of the algorithm is tested by altering the signal times there.

The processed dataset for this work is available at: <https://github.com/light2802/SUMO-Dataset>.

B. Resilience of Fog

The generated data is sent to a KNN-based machine-learning model periodically. KNN regression model is a non-parametric technique that, by averaging the data in the same

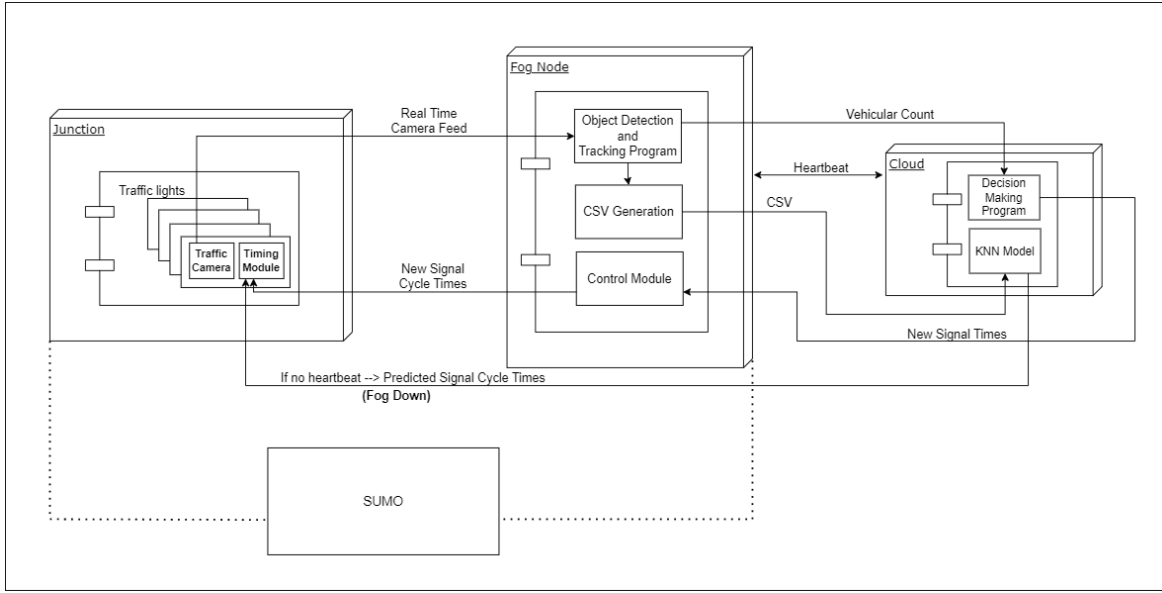


Fig. 1: Component diagram

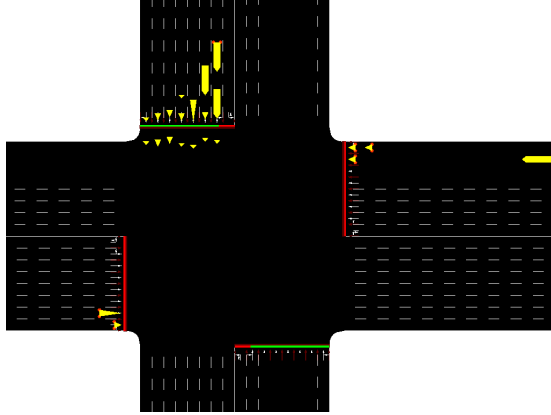


Fig. 2: SUMO Simulation

neighborhood, intuitively approximates the relationship between independent variables and the continuous result. This model runs on the Cloud. Based on historical data, it predicts the expected number of vehicles for the junctions under consideration. The distance equation used for this is as shown in Equation 1.

$$\text{dist}(x, z) = \left(\sum_{r=1}^d |x_r - z_r|^p \right)^{\frac{1}{p}} \quad (1)$$

The K value in the KNN algorithm defines the count of the nearest neighbors. Through trial and error, the optimal value of K for this use case is found to be 1.

In case of a Fog node failure causing an inability to collect live data, these predicted values are used to set the signal times for the duration one or more Fog nodes are down. Together with all other components it creates a quick, efficient, and resilient system to tackle the issue of road traffic congestion.

VI. RESULTS

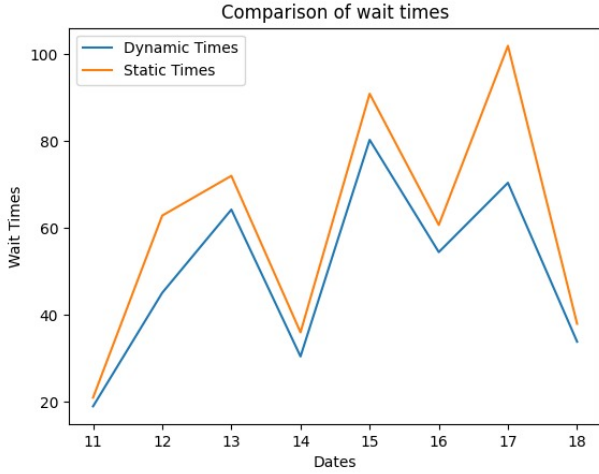
A. Comparison of Adaptive and Static Timings

The results obtained after simulating the case study in SUMO are summarized in this section. The data available is for the span of one week - 11th January 2023 to 18th January 2023. The metrics used for measurement are two inbuilt parameters available in SUMO. These are average wait time and average time loss. Wait Time (WT) indicates the time in which the speed of the vehicle was less than or equal to 0.1 m/s. Time Loss (TL) indicates the time that is lost because of driving below the ideal speed.

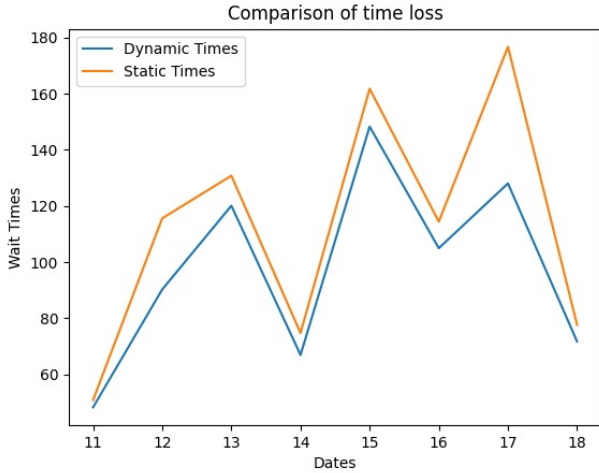
These values are used to check the validity of the algorithm and prove that the system does indeed reduce the wait times of vehicles leading to improved traffic flow. The graphs in Fig. 3 show that the adaptive traffic signal time cycle adjustment

Algorithm 1 Calculate optimal signal times

- 1: Set t to a list of length equal to the length of *vehicles*
 - 2: **for** each i in *vehicles* **do**
 - 3: Set *ratio* to i divided by *totalVehicles*
 - 4: Set *time* to *ratio* multiplied by *baseTimer*
 - 5: **if** *timeLimits*[0] is less than *time* and *time* is less than *timeLimits*[1] **then**
 - 6: Set $t[i]$ to *time*
 - 7: **else**
 - 8: Set $t[i]$ to the value in *timeLimits* (either *timeLimits*[0] or *timeLimits*[1]) that is closest to *time*
 - 9: **end if**
 - 10: **end for**
-



(a) Wait Times (in seconds)



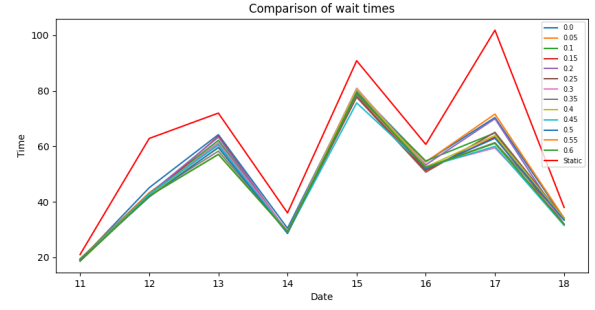
(b) Time Loss (in seconds)

Fig. 3: Comparison of metrics for static and dynamic times

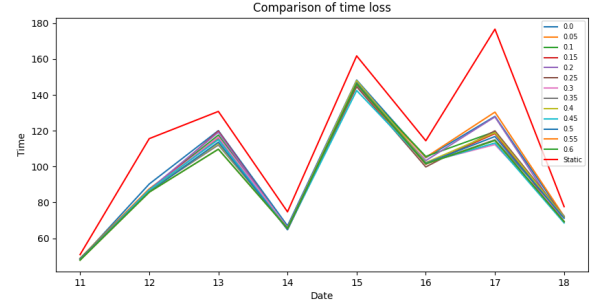
algorithm offers improved performance and reduces the wait time and time loss for vehicles. The differences between these metrics in static (S) and dynamic (D) traffic light times are shown in Table II. We observed a 17.624% reduction in waiting time and a 13.732% reduction in the time loss value.

B. Comparison of Statistics for Fog Resilience

To demonstrate the resilience of Fog nodes, some or all of the three Fog nodes are made to go down for a random amount of time. The simulation is run each time to produce downtime with probabilities ranging from 0.0 to 0.95 with increments of 0.05. The results are as shown in Fig. 4. The motive behind recording these values is to test whether the adaptive signal control system is resilient and performs better than the regular static system even under conditions of failure of one or more Fog nodes for a prolonged time. It is noticed that the metrics remain unchanged after a probability of 0.55. This is because, as the probability of failure increases, almost



(a) Wait Times (in seconds)



(b) Time Loss (in seconds)

Fig. 4: Comparative analysis of metrics for different probabilities of failure

all data is obtained from the predictions and almost none from the simulation. For the sake of brevity, the repeated values are omitted. Fig. 4 compares the average values of the two metrics for all probabilities for each day in consideration.

As is evident from the graphs, the system performs better than the static travel times even when the Fog nodes are down and real-time data is unavailable. The performance will enhance further as the model acquires more data to train on. We observe an average 21.271% and 16.764% reduction in wait time and time loss respectively.

C. Comparing Bandwidth Usage

Considering the use case of the city of Pune, we can estimate the bandwidth requirement for a purely Cloud-based system as opposed to a Cloud-plus Fog architecture.

The city of Pune has a total of 1068 CCTV cameras [15]. On average, one video requires a 3 Mbps capacity [16]. Thus the total bandwidth requirement for the city is about 3.4 Gbps. With the introduction of the Fog layer, every Fog node will require 12 Mbps capacity. Since only the count value is being sent to the Cloud, the value requiring 8 bytes for transmission, the bandwidth consumption goes down to 8.5 Kbps. This value needs to be transmitted once every 15 minutes. This further lowers the communication traffic. These values suggest more than 99% less bandwidth requirement for sending the data to Fog nodes as compared to streaming it to the Cloud.

TABLE II: Reduction in time metrics (in seconds)

Date	Average WT (S)	Average WT (D)	Average WT saved	Average TL (S)	Average TL (D)	Average TL saved
11/01/23	20.98	18.98	2	50.89	48.3	2.59
12/01/23	62.84	45.48	17.36	115.62	90.29	25.33
13/01/23	71.98	64.22	7.76	130.81	120.11	10.7
14/01/23	35.99	30.45	5.54	74.77	66.96	7.81
15/01/23	90.86	80.26	10.6	161.79	148.31	13.48
16/01/23	60.69	54.44	6.25	114.44	104.98	9.46
17/01/23	101.88	70.38	31.5	176.72	128.06	48.66
18/01/23	37.96	33.84	4.12	77.67	71.74	5.93
Average Values	60.3975	49.7563	10.6413	112.8388	97.4113	15.495

VII. CONCLUSION

The results obtained from this paper prove that using the capabilities of the Fog architecture in the creation of an adaptive traffic signal control system provides the following benefits:

- 1) Creation of a smart system that can process vast amounts of data in a distributed and efficient manner.
- 2) Decrease in average wait times of vehicles by 17.624% which indicates a decrease in traffic congestion.
- 3) Decrease in average time loss of vehicles by 13.732% which indicates increased vehicle speeds and improved traffic flow.
- 4) Creation of a resilient system that can perform better than the static method even under conditions of Fog node failure with a decrease in wait times by 21.271% and time loss by 16.764%.
- 5) Decrease in bandwidth utilization by 99.99% leading to an increase in efficiency and decrease in cost.
- 6) Providing data security and privacy by processing the video feed locally and only storing count values in the Cloud.

While the proposed solution offers promising results, some of the constraints of this paper are listed below:

- 1) The proposed solution makes use of data collected from only three junctions of the city, hence more data is required to test its performance on a city-wide scale.
- 2) Due to hardware constraints, the solution is tested on a simulation. The real-time implementation would provide more comprehensive results.
- 3) The object detection code for each camera feed coming from a specific camera depends on camera placement and the angle at which it faces the road. If a camera is angled in a way that bigger vehicles like trucks and buses cover the smaller vehicles behind it, the vehicular count will be less. This difference in the count would lead to less accurate predictions.

VIII. FUTURE SCOPE

This system can further be improved in the following aspects:

- 1) Addition of a feature to detect and give way to priority vehicles such as ambulances, fire trucks, or police cars. This can lead to quicker emergency response times,

increased safety, increased efficiency, and a better public image.

- 2) Detection and routing of ambulances to nearby hospitals can help emergency medical services have faster response times, enhanced patient care reduced delays.
- 3) Utilization of the collected data in urban planning and infrastructure development like improved infrastructure for pedestrians, cyclists, and public transportation.
- 4) Validate scalability of the proposed system: For the purpose of the paper, the data is available for only three junctions of the city for a span of one week. The proposed solution must be tested on a larger number of junctions with data acquired over a longer period of time.

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